

Q1: A uniform beam with a closed thin-walled cross section is shown in Figure 1. The beam is subjected to a torsional moment of $3500 \text{ N}\cdot\text{m}$ and is constrained to twist about the longitudinal axis passing through point C. For the curved wall 1–2, the thickness is 2 mm and the shear modulus is 20 GPa . For the other walls, the thickness and shear modulus are 1.6 mm and 25 GPa , respectively. Determine the rate of twist (radians per meter), the warping distribution, and the value of warping at points 1 through 4.

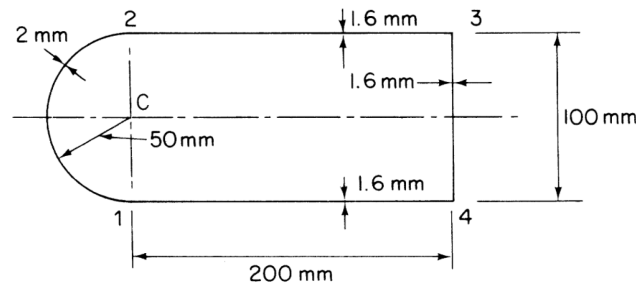


Figure 1: Q1's Cross-section under torsional torque

Rate of twist:

Warping distribution: (by symmetry, middle of 34, $w = 0$)

Warping of points 1 to 4:

Results:

$$\frac{d\theta}{dz} = 25.106 \times 10^{-6} \frac{\text{rad}}{\text{mm}}$$

$$w_s = 1.201 \times 10^{-3} \times \left(1.522 \oint_0^s ds - 4.179 \times 10^{-2} \times A_{os} \right)$$

$$w_3 = -w_4 = -0.159 \text{ mm}, w_2 = -w_1 = -0.0443 \text{ mm}, w_{21} = 0.0286 \phi - 0.0443 \text{ mm}$$

Q2: According to Figure 2, a beam with the shown cross section is subjected to a bending moment of $2000 \text{ N}\cdot\text{m}$ acting in its vertical plane. Determine the axial stress at points 1 through 8. ($t=4 \text{ mm}$)

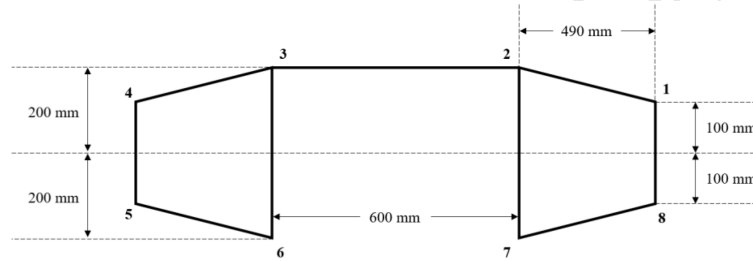
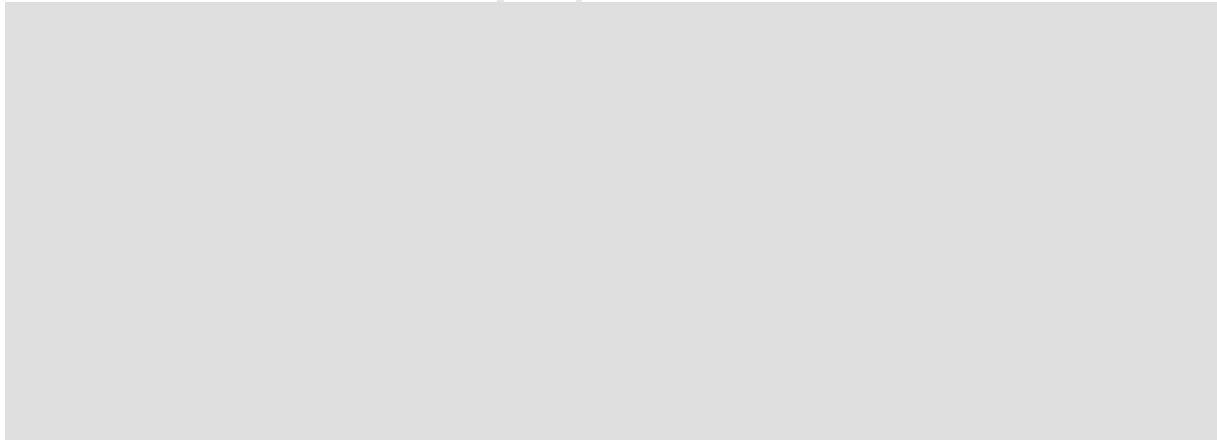


Figure 2: Q2's Beam cross-section under bending moment

Direct stress:



Results:

$$\sigma_1, \sigma_4 (y = 68.42 \text{ mm}) = 0.43 \frac{\text{N}}{\text{mm}^2}; \quad \sigma_2, \sigma_3 (y = 168.42 \text{ mm}) = 1.07 \frac{\text{N}}{\text{mm}^2}$$

$$\sigma_5, \sigma_8 (y = -131.58 \text{ mm}) = -0.83 \frac{\text{N}}{\text{mm}^2}; \quad \sigma_6, \sigma_7 (y = -231.58 \text{ mm}) = -1.47 \frac{\text{N}}{\text{mm}^2}$$

Q3: According to Figure 3, a beam with the shown cross section is subjected to a shear force of 100 kN applied in the vertical plane of symmetry. Calculate the shear stress at the midpoint of wall 3-6. (The thickness of all walls is 4 mm.)

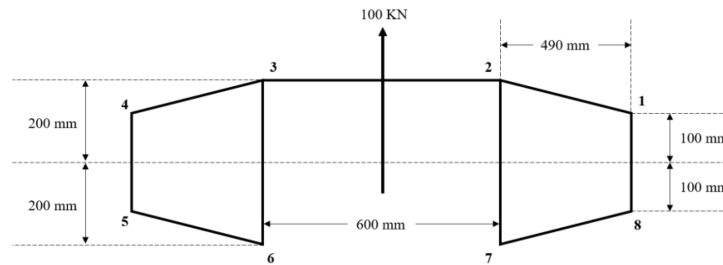


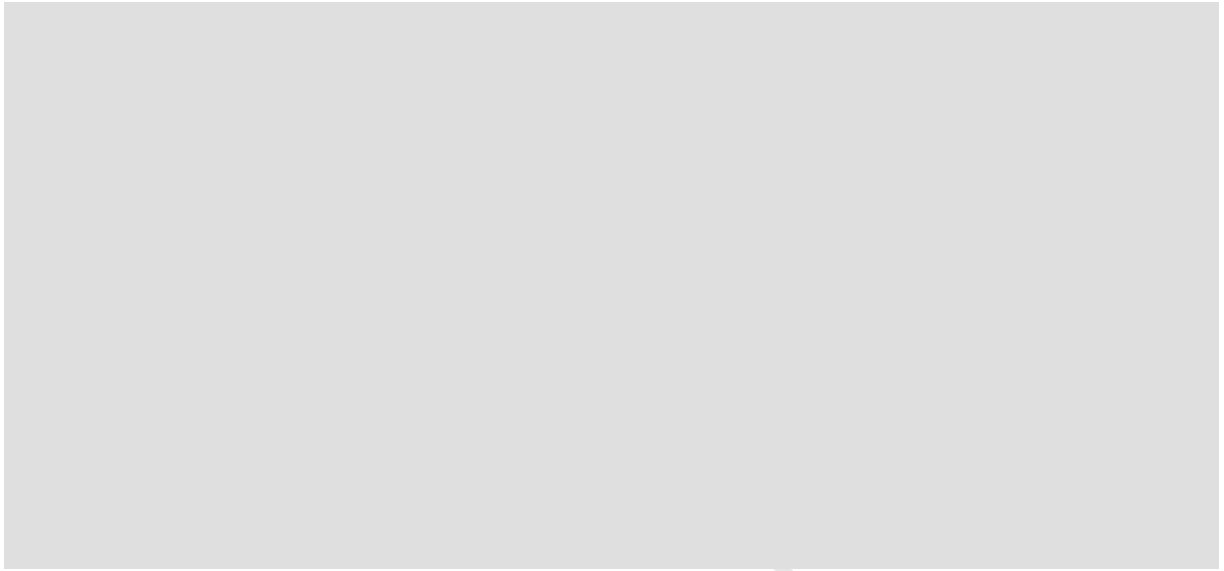
Figure 3: Q3's Beam cross-section under shear force

Shear stress:

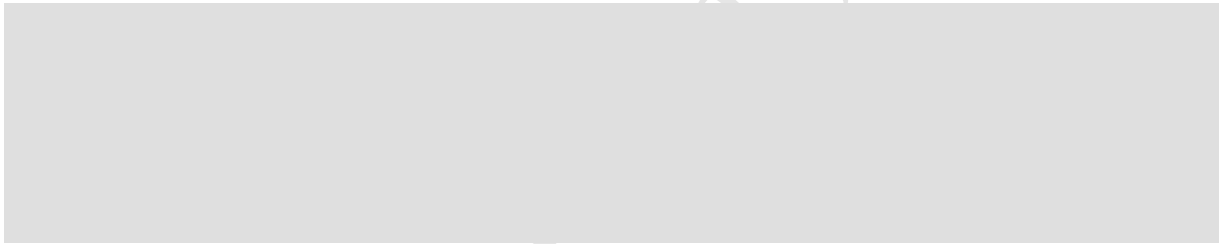
Name the middle of 32 as O:

Cut middle of the 18 and And number this point 9:

Calculate $q_{s,0}$:



substituting $q_{s,0}$ in $q_{b_{27}}$ equation:



Result:

$$\tau_{72_{mid}} = 55.33 \frac{\text{N}}{\text{mm}^2}$$

Q4: Idealize the cross section shown in Figure 4 as a combination of concentrated areas that carry axial normal stress at the four corners of the section, and panels that carry only shear stresses. Then, determine the distance of the shear center from the left wall.

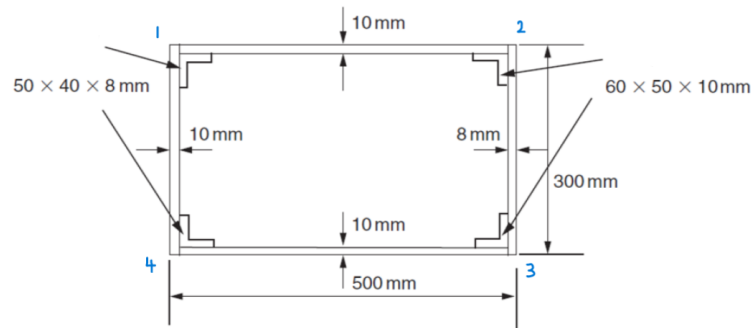


Figure 4: Q4's initial section for idealization

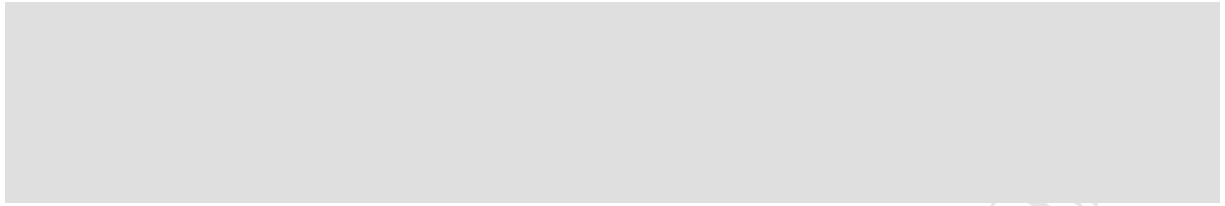
Boom area:

Shear stress:

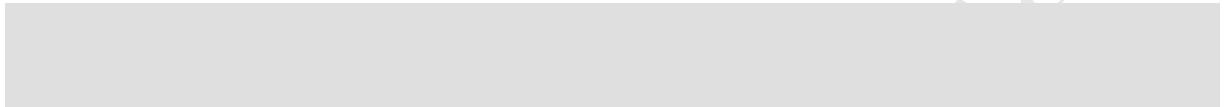
Cut 12:

When the shear force is applied at the SC, the twist becomes zero; then we have:

add $q_{s,0}$ to q_b values:



To find the distance, take moments about the midpoint of 1-4:



Results:

$$B_1 = B_4 = 3656 \text{ mm}^2, B_2 = B_3 = 3900 \text{ mm}^2$$

$$q_{12} = q_{34} = 0.0967 \times 10^{-3} S_y, q_{23} = -1.6273 \times 10^{-3} S_y, q_{41} = 1.7057 \times 10^{-3} S_y$$

$$\zeta = 229.59 \text{ mm}$$

Q5: Consider the idealized body shown in Figure 5 (having two axes of symmetry). The area of all booms is 200 mm^2 . If the cross section is subjected to a shear force of 50 kN and a bending moment of $100 \text{ kN}\cdot\text{m}$, determine the vertical normal stress in the booms (due to the bending moment) and the shear flow in the panels (due to the shear force).

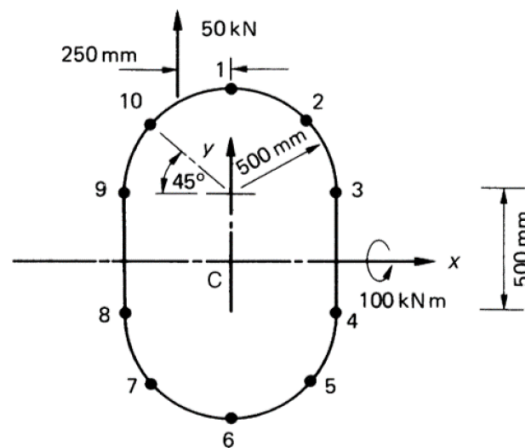


Figure 5: Q5's idealized cross-section under bending load and torsional moment

Direct stress:

Shear stress:

distances and I_{xx} :

Calculating stresses:

Calculating shears; cut 89:

To find $q_{s,0}$, take moments about point C. First calculate the areas:

Now calculate shear values:

Results:

$$\sigma_3 = \sigma_9 = -\sigma_4 = -\sigma_8 = 44.25 \frac{\text{N}}{\text{mm}^2}, \sigma_2 = \sigma_{10} = -\sigma_5 = -\sigma_7 = 106.83 \frac{\text{N}}{\text{mm}^2}$$

$$\sigma_1 = -\sigma_6 = 132.75 \frac{\text{N}}{\text{mm}^2}$$

$$q_{23} = q_{45} = -16.15 \frac{\text{N}}{\text{mm}}, q_{101} = q_{67} = 7.68 \frac{\text{N}}{\text{mm}}, q_{12} = q_{56} = -5.52 \frac{\text{N}}{\text{mm}}$$

$$q_{89} = -18.30 \frac{\text{N}}{\text{mm}}, q_{34} = -20.55 \frac{\text{N}}{\text{mm}}$$

Q6: Consider the idealized cross section subjected to a load of 44,500 N (Figure 6). The data for this cross section are provided in Table 1 and Table 2. Assuming that the shear modulus is constant over the entire cross section, determine the shear flow.

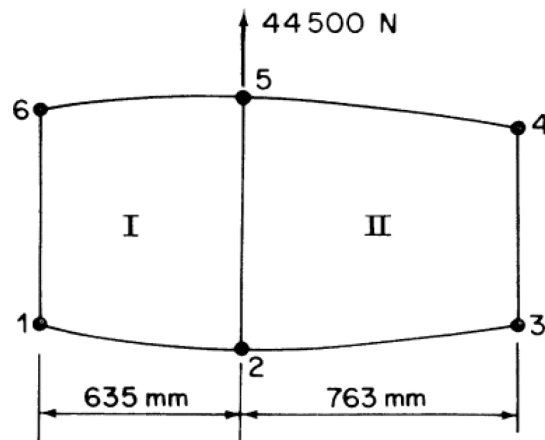
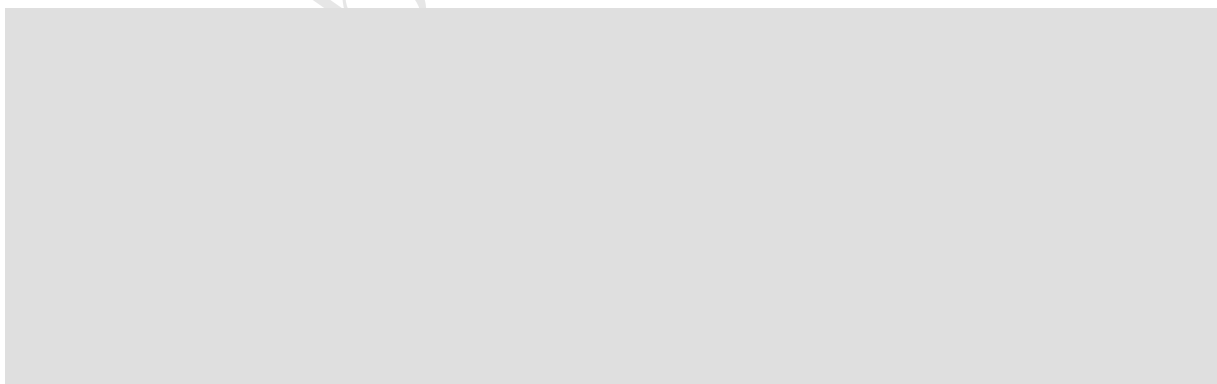


Figure 6: Q6's idealized cross-section under shear force

wall	l (mm)	t (mm)	Beam	Area (mm ²)
16	254	1.625	1, 6	1290
25	406	2.032	2, 5	1936
34	202	1.22	3, 4	645
56, 12	647	0.915	Cell I	232000
45, 23	775	0.559	Cell II	258000

Table 1: Section's data

Shear stress:



Cut 65 and 54 ans assume CCW q_{s0} for each cells:

Rate of twist:

We have $\delta = \int \frac{ds}{t}$, so:

For Cell I:

For Cell II:

The warping in the cells are equal:

Take moments about the midpoint of 25:

Solve two equations:

add $q_{s,0}$ values to q_b .

Results:

$$q_{56} = q_{12} = -0.78 \frac{\text{N}}{\text{mm}}, q_{54} = q_{23} = +7.31 \frac{\text{N}}{\text{mm}}$$

$$q_{61} = -34.79 \frac{\text{N}}{\text{mm}}, q_{43} = -20.83 \frac{\text{N}}{\text{mm}}$$

$$q_{52} = -81.59 + 0.78 + 7.31 = -73.5 \frac{\text{N}}{\text{mm}}$$

Q7: In Figure 7, assuming that the panels carry only shear stresses, determine the shear flow in the panels and the axial forces in the flanges and stiffeners.

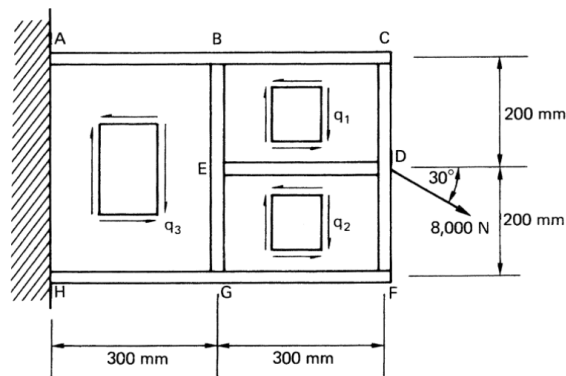


Figure 7: Q7's structure

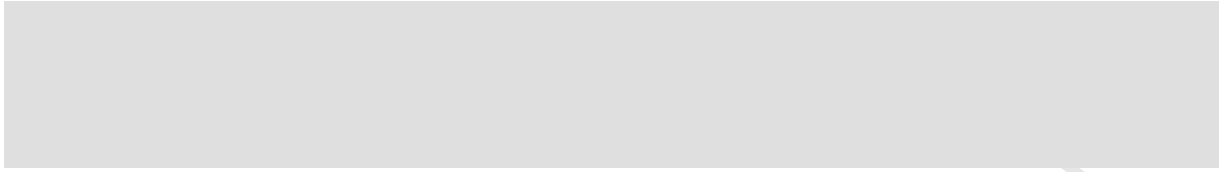
Write horizontal equilibrium for ED stiffener and vertical equilibrium for CDF stiffener:

ABEGH panel's vertical equilibrium:

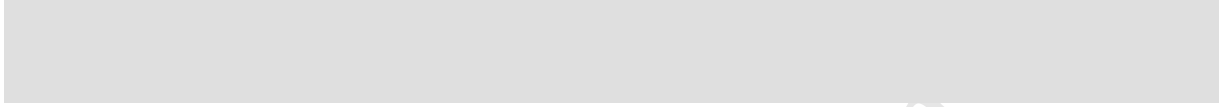
For axial forces in flanges and stiffeners:

Take section along $C \rightarrow B \rightarrow A$:

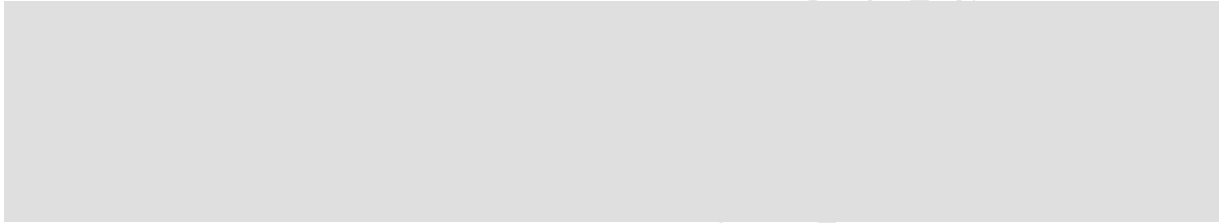
Take section along $H \rightarrow G \rightarrow F$:



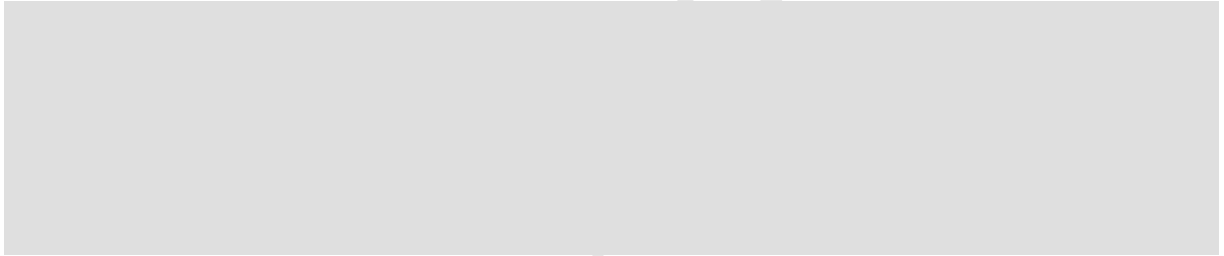
Take a section along $D \rightarrow E$:



Take section along $C \rightarrow D \rightarrow F$:



Take a section along $B \rightarrow E \rightarrow G$:



Results:

$$q_1 = 21.55 \frac{\text{N}}{\text{mm}}, q_2 = -1.55 \frac{\text{N}}{\text{mm}}, q_3 = 10 \frac{\text{N}}{\text{mm}}$$

$$P_{CB} = 21.55z_1, 0 \leq z_1 \leq 300 \text{ mm}, P_{BA} = 6465 + 10z_2, \text{ mm}(\text{horizontal})$$

$$P_{FG} = 1.55z_3, 0 \leq z_3 \leq 300 \text{ mm}, P_{GH} = 465 - 10z_4, 0 \leq z_4 \leq 300 \text{ mm}(\text{horizontal})$$

$$P_{DE} = -23.1z_5 + 6928.20, 0 \leq z_5 \leq 300 \text{ mm}(\text{horizontal})$$

$$P_{CD} = 21.55z_6, 0 \leq z_6 \leq 200 \text{ mm}, P_{DF} = -1.55z_7 + 310, 0 \leq z_7 \leq 200 \text{ mm}(\text{vertical})$$

$$P_{BE} = -11.55z_8, 0 \leq z_8 \leq 200 \text{ mm}, P_{EG} = 11.55z_9 - 2310, 0 \leq z_9 \leq 200 \text{ mm}(\text{vertical})$$

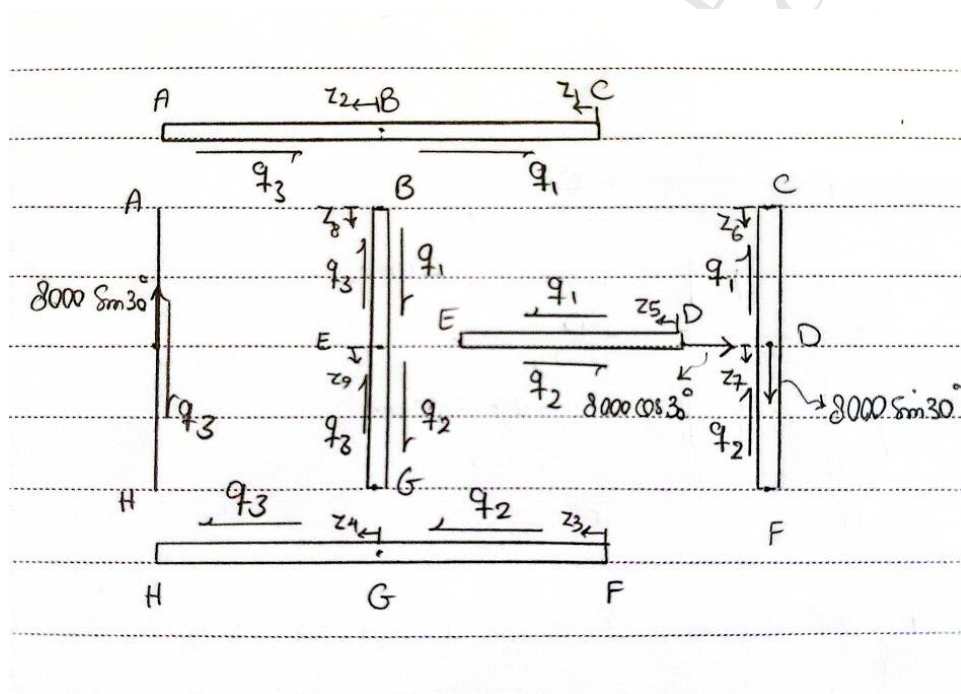


Figure 8: Q7's elements Decomposing